First, one has to be clear on the difference between $\supset$ and $\models$. The first is a connective in the object language. That is, inserting it between two sentences forms another sentence. The second is a relation in the metalanguage between two sentences (or, more generally, between a set of sentences and a sentence). You can read it as 'entails'. Note that 'the sun is red entails it is day' is not a grammatical sentence: it has three main verbs. Unfortunately, people often confuse $\supset$ and $\models$ because the first is often called 'implication', which it should not be. In classical logic, the connection between the two notions is this: $A \models B$ iff $\models A \supset B$.

Now, $\supset$ does not behave anything much like the English conditional. (For present purposes, think of this as being expressed by 'if'.) You suggest a different, three-valued connective, call this $\supset'$, as a better reading of 'if'. Exactly what this is is not completely specified. (What, for example, is the output if an input is I? You also do not say whether I is a designated value; I presume not.) Given what you do say, the connective does not have some of the weaknesses of the material conditional. For example, it does not satisfy $\neg A \models A \supset' B$. However, it would still have many others, e.g., $A, B \models A \supset' B$ (Trump is a narcissist, water is wet; so if Trump is a narcissist, water is wet.) And depending on how you fill in some of the other values, it may have new problems of its own. I think that the consensus of most professional logicians nowadays is that 'if' is not a truth function of any kind, two-valued or many-valued. Incidentally, none of this has anything to do with Explosion, which concerns $\models$, not $\supset$.

What you suggest cannot apply to $\models$ either. As any classical logician will tell you, this is not a truth function. That is, if I tell you the truth values of A and B, there is little you can tell me about the truth or otherwise of $A \models B$. Indeed, it's not a connective at all (so it doesn't make much sense to embed it in truth functions, such as $\neg$). And if you really insist that $A \models B$ is not true if A is false, what you are saying is that nothing validly follows from a false premise. How can that be, since we reason, apparently correctly, from false premises all the time? E.g.: Pluto is a planet. Therefore there are planets outside the orbit of Neptune.